Instituto de Física da USP

Report

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Summary

General Tasks Status

Task	Status	
Correlation Inference Study - Classical Approach		
Correlation Inference Study - Quantum Learning Approach		
2026-11 Paper		
2026-12 LASNPA Presentation		

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
$\hookrightarrow$ Uncertainty correlation effect	✓	Plots
RHIC Au-Au/d-Au Studies JETSCAPE		
$\hookrightarrow$ Uncertainty correlation effect	✓	Plots
Atlas ins1759506 studies		
$\hookrightarrow$ Uncertainty correlation inference	◔	Notebook
Synthetic data studies		
$\hookrightarrow$ Uncertainty correlation inference	○	

Atlas ins1759506 studies

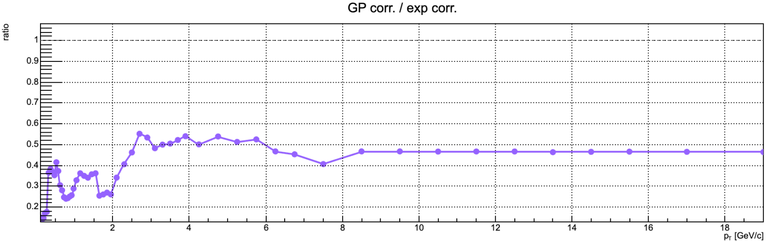

Figure: GP fit: correlation inference study

Atlas ins1759506 studies

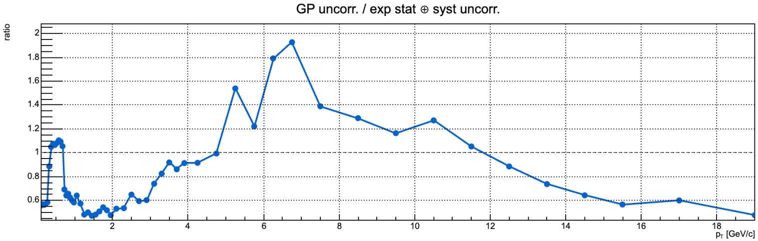

Figure: GP fit: correlation inference study

Atlas ins1759506 studies

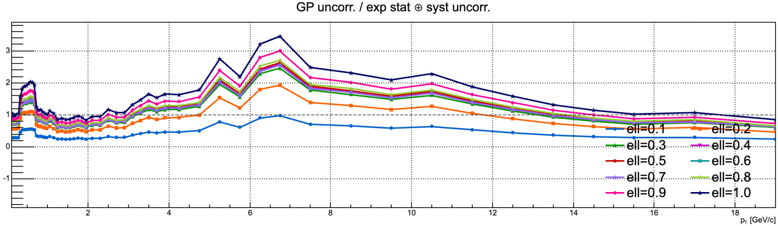

Figure: GP fit: correlation inference study

Selected Article

Taming systematic uncertainties at the LHC with the central limit theorem

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Abstract

We study the simplifications occurring in any likelihood function in the presence of a large number of small systematic uncertainties. We find that the marginalisation of these uncertainties can be done analytically by means of second-order error propagation, error combination, the Lyapunov central limit theorem, and under mild approximations which are typically satisfied for LHC likelihoods. The outcomes of this analysis are *i*) a very light treatment of systematic uncertainties *ii*) a convenient way of reporting the main effects of systematic uncertainties, such as the detector effects occurring in LHC measurements.

Figure: Simplifications in the likelihood and a convenient way to present the main effect of systematic uncertainties.